

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN

COURSE CODE: CSA1104

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4, BL5)
Assessment Tool Weightage: 25%

QUESTIONS: 05

Total Marks:25

Annexure A

Error Analysis Task

Code of Conduct:

I R. DURGA SREE / 192411189 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. DURGA SREE

RUBRICS FOR CALCULATION:

Error Analysis Task

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Error Identification	20%	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30%	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25%	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15%	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10%	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Error Analysis Task

Q1. Scenario: Online Banking System

Modules:

User Login, Fund Transfer, Account Verification

Question:

Write pseudocode for a quality assurance testing process that validates secure user authentication and fund transfer operations. Include input validation, transaction verification, exception handling, and test result logging.

PSEUDOCODE:

START

LOGIN:

Enter username and password

Validate input fields

IF fields are empty THEN

 Log "Invalid login input"

 Display error message

ELSE

 Authenticate user

 IF authentication fails THEN

 Log "Login failed"

 Display "Invalid credentials"

 ELSE

Log "Login successful"

FUND TRANSFER:

Enter receiver account, amount and PIN

Validate account number and amount

IF input is invalid OR amount ≤ 0 THEN

Log "Invalid transfer request"

Display error message

ELSE

Verify sender account and balance

IF account verification fails THEN

Log "Account verification failed"

ELSE IF balance $<$ amount THEN

Log "Insufficient balance"

Display error message

ELSE

Process transaction

IF transaction successful THEN

Verify transaction ID and transferred amount

Log "Transfer successful"

ELSE

Log "Transfer failed"

Rollback transaction

END IF

END IF

END IF

EXCEPTION:

IF system/network/database error occurs THEN

Rollback incomplete transaction

Log exception details

Display "Transaction could not be completed"

Generate test report

END

Q2. Scenario: E-Commerce Shopping Platform

Features:

Product Search, Shopping Cart, Payment Gateway

Question:

Design pseudocode for a usability testing evaluation that measures customer interaction efficiency, including product search time, checkout completion time, and navigation complexity. Explain how the algorithm collects and analyzes user behavior data.

PSEUDOCODE:

START

Initialize test user

Record start time

PRODUCT SEARCH:

User enters product name

Record search start time

Display search results

Record search end time

Calculate SearchTime

SHOPPING CART:

User selects product

Add product to cart

Record navigation steps

Measure time taken to reach cart

CHECKOUT:

User proceeds to checkout

Record checkout start time

Enter required details

Complete payment

Record checkout end time

Calculate CheckoutTime

USABILITY ANALYSIS:

Calculate average SearchTime

Calculate average CheckoutTime

Count number of navigation steps

Count user errors

Record failed interactions

IF SearchTime is high THEN

 Mark "Search needs improvement"

IF CheckoutTime is high THEN

 Mark "Checkout needs improvement"

IF navigation steps are excessive THEN

 Mark "Navigation is complex"

Generate usability report

Store user behavior data and results

END

Q3. Scenario: Hospital Management System

Modules:

Patient Registration, Appointment Scheduling, Medical Records

Question:

Write pseudocode to generate and execute test cases for appointment scheduling functionality. Include validation of doctor availability, appointment conflicts, patient information accuracy, and system responses to invalid requests.

PSEUDOCODE:

START

Generate test cases:

TC1 = Valid appointment

TC2 = Invalid patient information

TC3 = Doctor unavailable

TC4 = Conflicting appointment

TC5 = Invalid date/time

TC6 = Missing required information

FOR each test case DO

Enter patient information

Enter doctor and appointment details

Validate patient information

IF information is invalid THEN

Display error

Log test as PASS if error is correctly detected

CONTINUE

END IF

Check doctor availability

IF doctor is unavailable THEN

Display "Doctor unavailable"

Log result

CONTINUE

END IF

Check appointment conflicts

IF conflict exists THEN

Display "Appointment conflict"

Log result

CONTINUE

END IF

Validate date and time

IF date/time is invalid THEN

 Display error

 Log result

 CONTINUE

END IF

Create appointment

IF appointment created successfully THEN

 Verify appointment details

 Log test as PASS

ELSE

 Log test as FAIL

END IF

END FOR

Generate test execution report

END

Q4. Scenario: Airline Reservation System

Modules:

Passenger Registration, Flight Booking, Seat Allocation

Question:

Develop pseudocode for a constraint-based testing approach where booking requests must satisfy conditions such as valid passenger information, seat availability, and payment confirmation. Show how invalid booking scenarios are detected and handled.

PSEUDOCODE:

START

Receive booking request

Validate passenger information

IF name, age, passport or contact details are invalid THEN

 Reject booking

 Log "Invalid passenger information"

ELSE

 Check flight availability

 IF flight does not exist THEN

 Reject booking

 Log "Invalid flight"

 ELSE

 Check seat availability

 IF no suitable seat is available THEN

 Reject booking

 Log "Seat unavailable"

 ELSE

 Select seat

 Verify seat is not already allocated

 IF seat is already allocated THEN

Reject booking

Log "Seat allocation conflict"

ELSE

Process payment

IF payment is not confirmed THEN

Cancel booking

Release selected seat

Log "Payment failed"

ELSE

Confirm booking

Allocate seat

Generate booking reference

Log "Booking successful"

END IF

END IF

END IF

END IF

END IF

Handle unexpected exceptions:

Rollback incomplete booking

Release reserved seat

Log exception

Generate booking test report

END

Q5. Scenario: Smart Library Management System

Features:

Book Search, Book Reservation, Borrow/Return Services

Question:

Write pseudocode for integrating quality assurance and usability testing where the system evaluates both functional correctness (reservation and borrowing operations) and user experience factors such as response time, ease of navigation, and accessibility.

PSEUDOCODE:

START

Initialize testing system

BOOK SEARCH:

Enter book title/author

Start timer

Display search results

Stop timer

Calculate ResponseTime

Verify search results are correct

BOOK RESERVATION:

Select available book

Click Reserve

Verify reservation status

IF book is unavailable THEN
 Display appropriate message
 Log result

ELSE
 Confirm reservation
 Log reservation result

END IF

BORROW SERVICE:

 Select reserved/available book
 Verify member details
 Process borrowing request
 IF borrowing rules are satisfied THEN
 Update book status to "Borrowed"
 Log successful borrowing

 ELSE
 Display error
 Log failed borrowing

 END IF

RETURN SERVICE:

 Return borrowed book
 Update book status to "Available"
 Calculate fine if applicable

Log return operation

USABILITY TEST:

Measure page response time

Count navigation steps

Check button and menu accessibility

Check readability and keyboard navigation

Record user errors

ANALYZE RESULTS:

Calculate average response time

Calculate task completion rate

Calculate navigation complexity

Identify accessibility problems

Identify functional errors

Generate QA and usability report

END